Berwickshire High School

Physics Department

S3 Physics • Electricity 1: Circuits and Resistance

Practice Test

Name: ______________________________ Date: _________________

Answer all questions. 20 marks. 30 minutes.

You may use the resistor colour-band reference sheet and the N5 relationships sheet.

Score	Percentage	Grade
/20	%	

Section A

Each question is worth 1 mark. Circle the letter of the correct answer.

1. **1**

Four circuit symbols, W, X, Y and Z, are shown. Which row identifies the components represented by these symbols?

A W ammeter, X lamp, Y resistor, Z fuse

B W voltmeter, X lamp, Y fuse, Z resistor

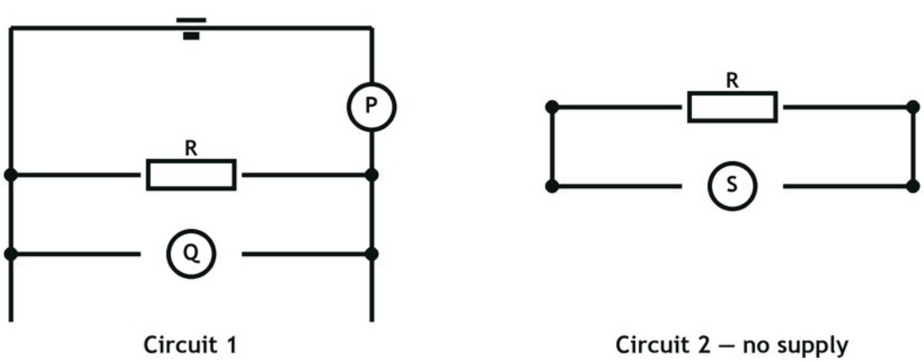
C W ammeter, X lamp, Y fuse, Z resistor

D W voltmeter, X motor, Y resistor, Z fuse

E W voltmeter, X lamp, Y resistor, Z fuse

2. **1**

Two circuits are used to determine the resistance of resistor R, as shown. Which row in the table identifies meter P, meter Q and meter S?



A P voltmeter, Q ammeter, S ohmmeter

B P ohmmeter, Q voltmeter, S ammeter

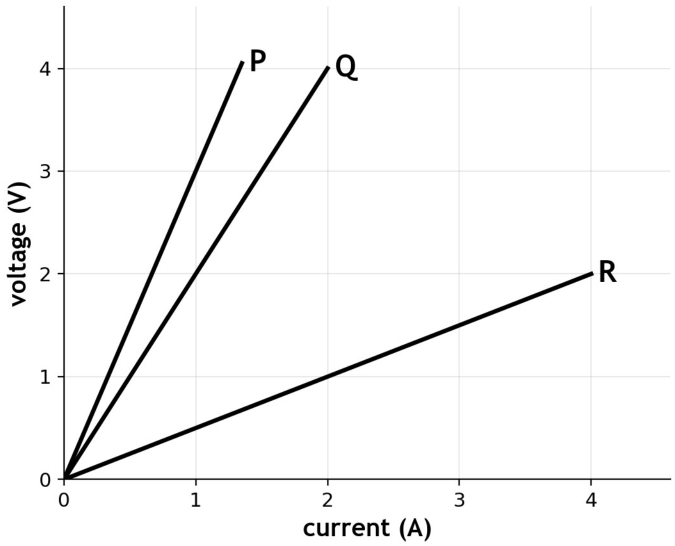
C P ammeter, Q voltmeter, S ohmmeter

D P ammeter, Q ohmmeter, S voltmeter

E P voltmeter, Q ohmmeter, S ammeter

3. **1**

The graph shows how the voltage varies with current for three resistors P, Q and R. A student makes the following statements. I The resistance of resistor P is greater than that of resistors Q and R. **II** When the voltage across resistor Q is 2.0 V, the current in the resistor is 1.0 A. **III** The resistance of resistor R is 2.0 Ω. Which of these statements is/are correct?



A I only

B II only

C I and II only

D II and III only

E I, II and III

4. **1**

The current in a circuit is a measure of the

A energy given to each coulomb of charge

B charge passing a point each second

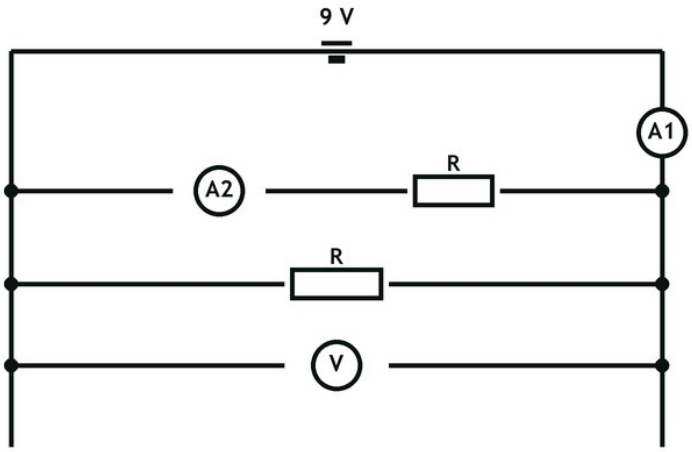
C speed of the charges in the wire

D resistance of the circuit

E voltage across the supply

5. **1**

A circuit is set up as shown. The resistors are identical. Which row in the table shows the reading on the voltmeter and possible readings on ammeters A1 and A2?



A 4.5 V, 0.4 A, 0.2 A

B 9 V, 0.2 A, 0.4 A

C 9 V, 0.4 A, 0.2 A

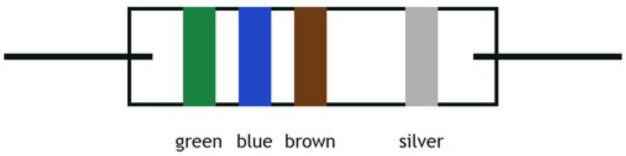
D 4.5 V, 0.2 A, 0.2 A

E 9 V, 0.2 A, 0.2 A

Section B

6.

The diagram below shows the colour coding for a resistor. Using the information in the resistor colour-band reference sheet:



(a) Determine the resistance of the resistor. **1**

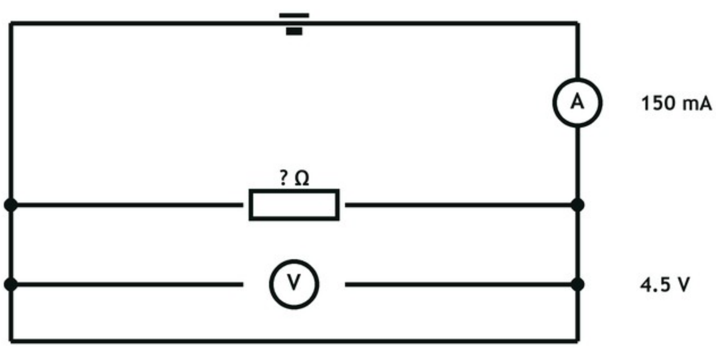
(b) State the tolerance in the resistance of the resistor. **1**

(c) Determine the maximum value of resistance of the resistor. **1**

(d) Determine the minimum value of resistance of the resistor. **1**

7.

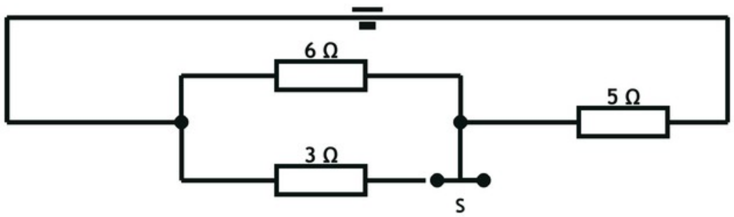
A student sets up the circuit as shown.



Calculate the resistance of the resistor. **3**

8.

A circuit is set up as shown.



Determine the total resistance of the circuit when switch S is closed. **4**

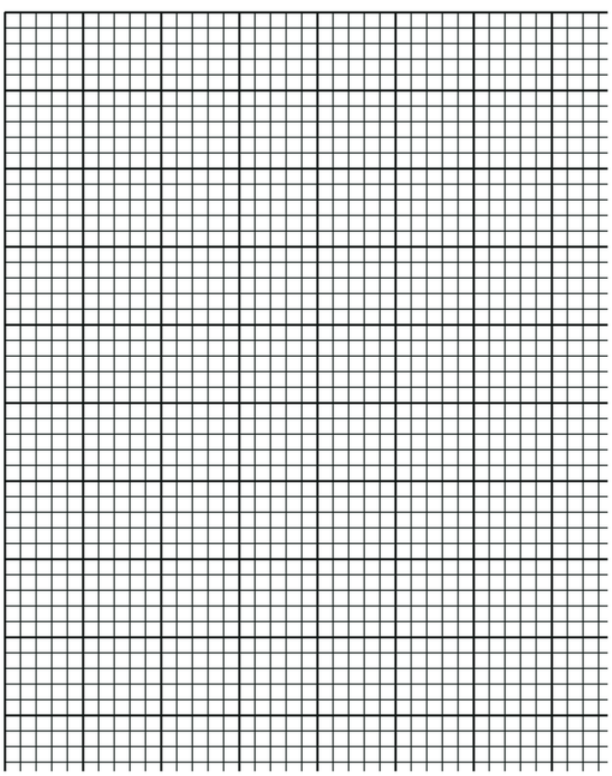
9.

A student investigates the relationship between the voltage across a lamp and the current in it, and records the following values.

Voltage (V)	0.00	0.50	1.00	1.50	2.00
Current (mA)	0	120	200	250	280

(a) On the graph paper provided, plot a graph of the results. **3**

(b) Describe what happens to the resistance of the lamp as its temperature increases. **1**



(end of question paper)

S3 Electricity 1 — Practice Test
Marking Instructions

Section A

1 B · 2 C · 3 C · 4 B · 5 C

Section B

Q6 — Resistor colour code (4)

(a) Green = 5, Blue = 6 → 56; Brown = ×10 → 560 Ω 1

(b) Silver = ±10% 1

(c) 10% of 560 = 56 Ω; 560 + 56 = 616 Ω 1

(d) 560 − 56 = 504 Ω 1

Accept 0.56 kΩ / 0.616 kΩ / 0.504 kΩ. (c) and (d) carry through an incorrect (a).

Q7 — Ohm's law from a circuit (3)

Converts 150 mA to 0.150 A 1

Substitutes into R = V ÷ I: 4.5 ÷ 0.150 1

R = 30 Ω (unit required) 1

A candidate who does not convert (4.5 ÷ 150 = 0.03 Ω) scores 1 mark only.

Q8 — Combination circuit (4)

Identifies 6 Ω and 3 Ω as parallel, correct method 1

Parallel combination = 2 Ω 1

Adds the 5 Ω in series 1

Total = 7 Ω (unit required) 1

(6×3)÷(6+3) = 2 Ω; 2 + 5 = 7 Ω. 6+3+5=14 Ω (treats all as series) scores 0.

Q9 — Plotting and temperature (4)

(a) Axes labelled with quantity + unit, sensible scale 1

All five points plotted accurately 1

Best-fit curve drawn (smooth, not dot-to-dot) 1

(b) Resistance increases 1

Either axis orientation accepted (V up/I across, or I up/V across) provided both axes carry quantity and unit. Accept the current axis in mA or A.

Totals: 5 + 4 + 3 + 4 + 4 = 20